Smart meter consumption data: Data quality report

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Data availability

The following tables summarise the amount of valid data among participants recruited in waves 1, 2 and 3. For more information about error definitions, see the Errors section below or the *serl_smart_meter_documentation_edition07.pdf* accompanying document. The data are presented as percentages on the next page. The abbreviation 'HH' stands for 'half-hourly'.

Since September 2022 the smart meter data collection has moved from UKDS to the Advanced Research Computing Center (ARCC) at UCL. This system migration required several changes to the underlying methods of data collection, and there have been some minor impacts to the data quality. Very little electricity active export data was collected in September, October and November 2022 and February 2023 while some system migration activities were conducted because this data is only stored on the meter for 3 months and we were unable to backfill missing data once migration

activities were completed. The same months have low levels of reactive import and export data for the same reason. The data quality for half hourly electricity import readings has reduced with a greater level of missing data. Daily electricity import data quality is reduced for September and October 2022 and April to June 2023 with a greater number of invalid read times (i.e. read time is not midnight) during these periods. This may be associated with the BST time zone during these periods and is something we will investigate. As always, feedback on the SERL data or data quality is greatly appreciated.

Additionally, since Edition05 we have increased the minimum threshold for flagging 'high' electricity active export readings, as domestic solar panel arrays have increased in size and some households are now exporting more than our original 4kW cut-off. Therefore a few export reads previously flagged as errors will now be marked valid in this edition.

Smart meter consumption data: Technical documentation describes changes that have been made to the time stamp for the daily data and to the 'Valid read but invalid read time 'daily flag. The impact of these changes are noticeable if the tables and figures in this report are compared with those for Edition06:

- Tables 1 and 2 show a higher proportion of high percentages valid daily reads for both electricity and gas because of the increase in reads defined as having a valid timestamp (daily reads at 00:00 UTC during British Summer time are now defined as valid).
- Figure 1 shows improved electricity daily data availability in September and October 2022 and during BST in 2023, as more daily readings were defined as valid.
- Table 6 shows a lower figure for 'Valid read but invalid read time' because of the changed definition of valid read time.

The first row of Table 1 shows the number of participants for which the read type above was possible. Over 4,500 participants have at least 90% of all (non-reactive) import readings valid and nearly 700 with at least 75% of all export readings valid. Some participants have "read type possible" for half-hourly data but not for daily. This is either because these participants withdrew consent after we started collecting half-hourly data but before daily reading collection began, or because they have a SMETS1 meter that doesn't collect daily reads.

Note that where daily readings are invalid or unavailable it is often possible to impute the reading with the sum of the half-hourly readings for that day. Where possible, a valid half-hourly sum is provided for comparison/imputation with the daily reads.

Table 1: Number of participants with different minimum levels of valid import ('im') and export ('ex') readings.

	Daily gas (im)	*HH gas (im)*	*Daily elec (active im)*	*HH elec (active im)*	*HH elec (reactive im)*	*HH elec (active ex)*	*HH elec (reactive ex)*
Read type possible	9,333	10,114	12,241	13,208	13,208	804	804
100% valid	2,113	1,041	1,341	2,757	49	1	1
>= 99% valid	4,303	4,350	3,940	5,659	80	7	7
>= 98% valid	4,810	5,308	4,294	6,955	152	13	13
>= 95% valid	5,428	6,351	4,848	7,995	309	71	71
>= 90% valid	5,873	6,945	5,324	8,533	4,481	519	519
>= 75% valid	6,818	7,555	7,409	9,088	10,099	697	697
>= 50% valid	7,558	8,382	8,822	11,359	11,237	738	738
>= 25% valid	8,315	9,421	10,653	12,550	12,261	787	787
>= 10% valid	8,532	9,757	11,026	12,831	12,835	797	797
>= 5% valid	8,583	9,848	11,186	12,938	13,007	802	802
>= 0% valid	9,333	10,114	12,241	13,208	13,208	804	804

Table 2 shows the data from Table 1 presented as percentages. The first row of data ('Read type possible') shows the percent of all participants for which we expect to have each type of read. Approximately 77% of households have a gas meter registered with the DCC (i.e. a SMETS2 gas meter or an upgraded SMETS1 meter). The percentages in the rows that follow are percentage of valid readings from those meters for which we expect the read type, e.g. 64.6% of households with a smart meter that we could collect data from have at least 90% of their half-hourly electricity active import readings available and valid. 6.1% of participants have at least 1 export read.

Table 2: Percentage of participants with different minimum levels of valid import ('im') and export ('ex') readings. 'Read type possible' indicates the percentage of participants with that type of read available.

	Daily gas (im) (%)	*HH gas (im) (%)*	*Daily elec (active im) (%)*	*HH elec (active im) (%)*	*HH elec (reactive im) (%)*	*HH elec (active ex) (%)*	*HH elec (reactive ex) (%)*
Read type possible	70.7	76.6	92.7	100.0	100.0	6.1	6.1
100% valid	22.6	10.3	11.0	20.9	0.4	0.1	0.1
>= 99% valid	46.1	43.0	32.2	42.8	0.6	0.9	0.9
>= 98% valid	51.5	52.5	35.1	52.7	1.2	1.6	1.6
>= 95% valid	58.2	62.8	39.6	60.5	2.3	8.8	8.8
>= 90% valid	62.9	68.7	43.5	64.6	33.9	64.6	64.6
>= 75% valid	73.1	74.7	60.5	68.8	76.5	86.7	86.7
>= 50% valid	81.0	82.9	72.1	86.0	85.1	91.8	91.8
>= 25% valid	89.1	93.1	87.0	95.0	92.8	97.9	97.9
>= 10% valid	91.4	96.5	90.1	97.1	97.2	99.1	99.1
>= 5% valid	92.0	97.4	91.4	98.0	98.5	99.8	99.8
>= 0% valid	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Number of participants with data over time

The following figure shows how many participants have at least 75% of each month's data available and flagged as valid. It should be noted that the definition of valid is different to that in tables 1 and 2; in the chart daily valid includes days with a valid sum of half-hourly reads where the daily read is not valid/available. A participant's earliest read date depends on the date they signed up (we can collect up to 1 year of historic data from consent date (3 months for export data)), when their SMETS2 meter was installed (or upgraded if SMETS1), and when they moved into the house (if they moved in within the year preceding sign up).

For daily data we see a steady increase in the numbers from September 2018 (1 year before the first sign up) until January 2021, reflecting the availability of historic data for participants recruited in different waves. Half-hourly data collection dates back to the start of 2019 and increases in a similar way to the daily data. The half-hourly reactive import and export data is not shown on this figure, as we believe very few researchers are using this data stream. The half-hourly reactive import and export data is only stored on the meter for 3 months and so data availability increases sharply around 3 months before each recruitment wave.

The slow reduction from the start of 2021 reflects the fact that all historic data for participants has been collected and collection is now in ongoing/longitudinal mode, with a small number of participants moving away or withdrawing consent each month and no new participant recruitment. As described above, the technical and operational issues collecting electricity data in the latter part of 2022 and 2023 relating to system migration activities are visible in this figure.

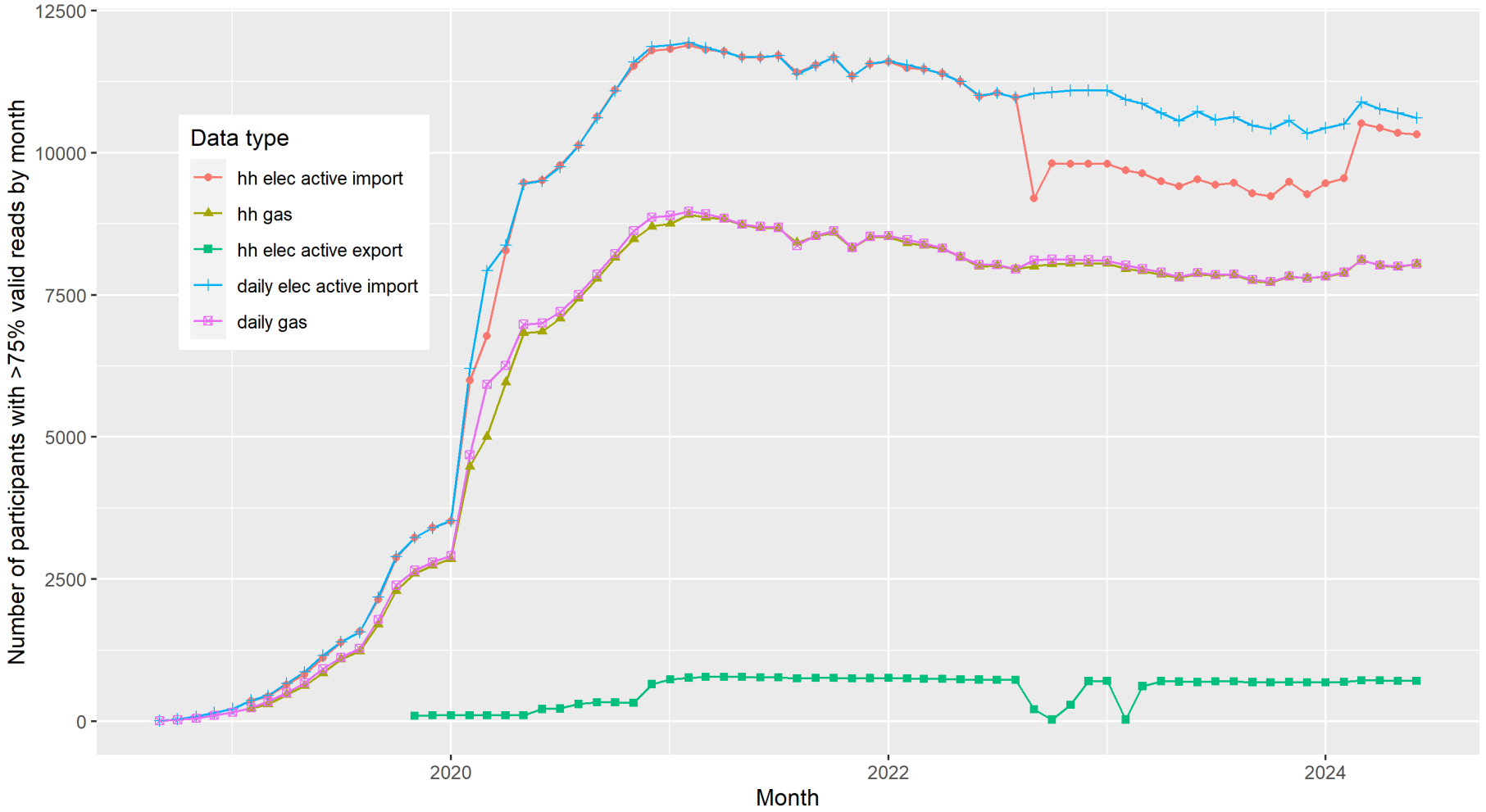


Figure 1: Number of participants in the SERL Observatory with at least 75% data valid per month. Daily valid includes days with a valid sum of half-hourly reads where the daily read is not valid/available.

Errors

More details about the error flags/codes can be found in the *serl_smart_meter_documentation_edition07.pdf* accompanying document.

Half-hourly data

The half-hourly data runs from 2019-01-13 to 2024-07-01 and 13142 participants have half-hourly reads. For each half-hourly read, for each participant, the following tables show the number of errors and valid reads in the half-hourly electricity and gas datasets. Abbreviations: 'act' (active), 'react' (reactive) and 'elec' (electricity).

Note that all 13209 participants have an electricity meter and 3252 participants don't have a gas meter (24.6%). The main issue for half-hourly electricity active import and export reads is missing data.

Table 3: Half-hourly electricity active import and export readings: number and percent of readings with each error flag.

Flag	*Meaning*	*Act elec import (N)*	*Act elec import (%)*	*Act elec export (N)*	*Act elec export (%)*
1	Valid	842,286,935	82.8	46,801,050	84.9
0	Missing	174,768,363	17.2	8,323,413	15.1
-1	Max read	661	0.0	11	0.0
-2	Very high but not max	877	0.0	36	0.0
-3	Negative	0	0.0	0	0.0
-4	Elec in kWh	0	0.0	0	0.0
-5	Valid read but invalid read time	332	0.0	34	0.0
-6	Suspicious zero	0	0.0	0	0.0

The electricity reactive import and export reads mainly have issues with missing data.

Table 4: Half-hourly electricity reactive import and export readings: number and percent of readings with each error flag.

Flag	*Meaning*	*React elec import (N)*	*React elec import (%)*	*React elec export (N)*	*React elec export (%)*
1	Valid	699,230,245	77.5	46,801,069	84.9
0	Missing	202,530,899	22.5	8,323,413	15.1
-1	Max read	60	0.0	0	0.0
-2	Very high but not max	29	0.0	28	0.0
-3	Negative	0	0.0	0	0.0
-4	Elec in kWh	0	0.0	0	0.0
-5	Valid read but invalid read time	367	0.0	34	0.0
-6	Suspicious zero	0	0.0	0	0.0

The half-hourly gas reads have similar availability to the half-hourly electricity active import reads, but suffer from a few incredibly high readings. The 'max read' error flag (-1) is when the read is the largest number storable/transmittable (all 1s in binary). For more details see the aforementioned smart meter data documentation.

Table 5: Half-hourly gas import readings: number and percent of readings with each error flag.

Flag	*Meaning*	*Gas import (N)*	*Gas import (%)*
1	Valid	646,615,416	83.2
0	Missing	127,412,010	16.4
-1	Max read	3,019,614	0.4
-2	Very high but not max	8,857	0.0
-3	Negative	0	0.0
-4	Elec in kWh	0	0.0

Flag	Meaning	Gas import (N)	Gas import (%)
-5	Valid read but invalid read time	69,431	0.0
-6	Suspicious zero	0	0.0

Daily Data

The daily data runs from 2018-08-18 to 2024-06-30. A total of 13209 participants are included in these datasets. Of these, 13137 have either daily reads or valid half-hourly sums to substitute for a daily read. For each daily read, for each participant, the following table shows the number of errors and valid reads in the daily electricity and gas datasets. Note that in this table the gas and electricity reads are included together, as the only daily read types are gas import and electricity active import. There is greater availability of the gas data than the electricity data. Around 9% of the electricity data was suspected to be recorded in kWh not in Wh as prescribed by the Smart Energy Code. Around 2% of daily electricity readings are zero and flagged as suspicious; the vast majority of these readings are from meters which start recording zero for some or all of British Summer Time. See the aforementioned smart meter documentation for more details.

Table 6: Daily electricity active import readings and daily gas import readings: number and percent of readings with each error flag.

Flag	Meaning	Act elec import (N)	Act elec import (%)	Gas import (N)	Gas import (%)
1	Valid	14,042,241	70.8	12,013,655	79.6
0	Missing	4,052,467	20.4	2,905,667	19.3
-1	Max read	118	0.0	158,933	1.1
-2	Very high but not max	2,906	0.0	5,899	0.0
-3	Negative	0	0.0	0	0.0
-4	Elec in kWh	1,436,579	7.2	0	0.0
-5	Valid read but invalid read time	197	0.0	1,269	0.0

Flag	Meaning	Act elec import (N)	Act elec import (%)	Gas import (N)	Gas import (%)
-6	Suspicious zero	292,411	1.5	0	0.0

Further reading

More information can be found about the SERL data in our 2021 open access journal paper The SERL Observatory Dataset: Longitudinal Smart Meter Electricity and Gas Data, Survey, EPC and Climate data for over 13,000 Households in Great Britain. SERL Statistical Reports are made available via the SERL website.